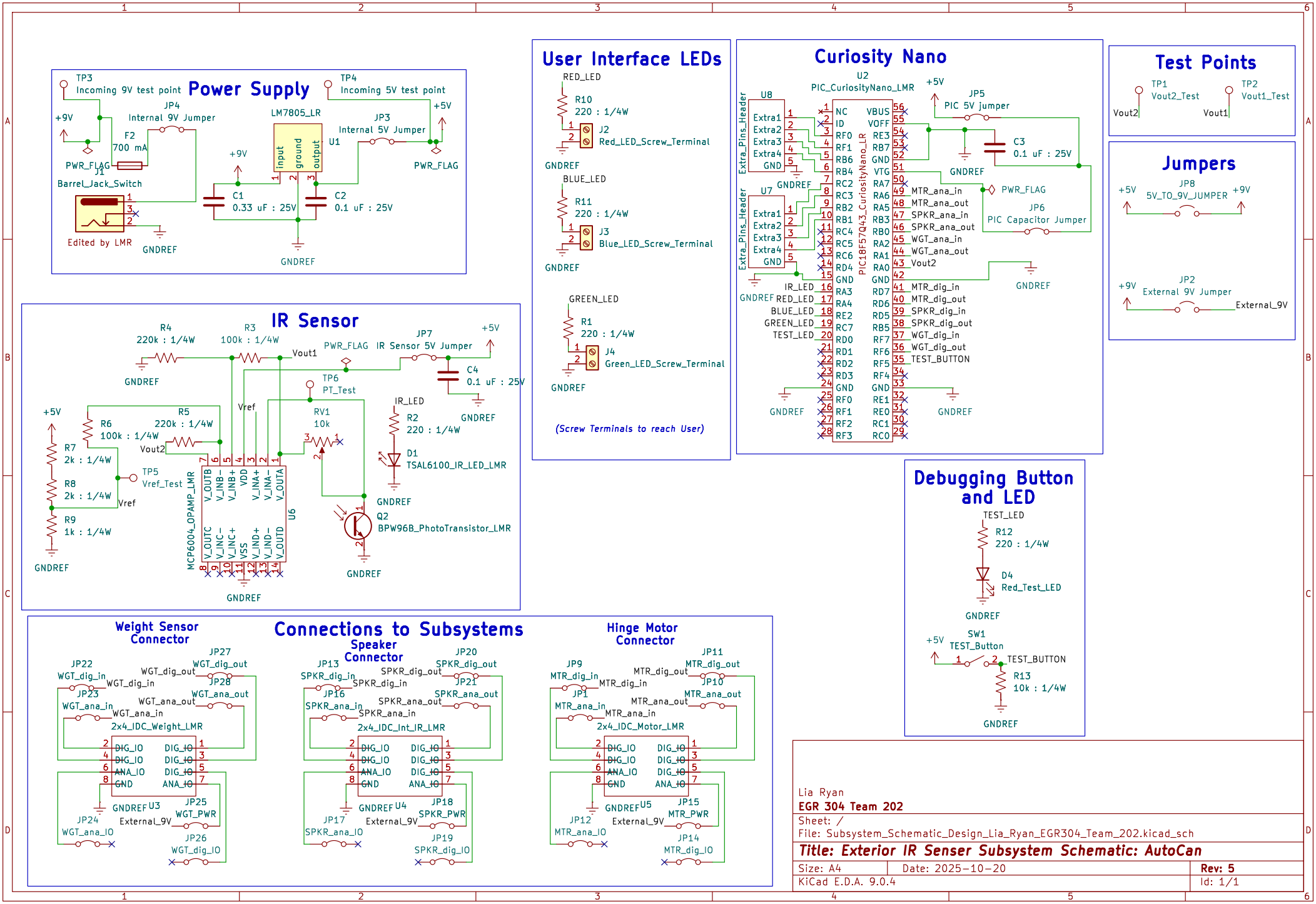
Power Supply
TP3
Incoming 9V test point
+9V
JP4
Internal 9V Jumper
F2
700 mA
PWR_FLAG
J1
Barrel_Jack_Switch
Edited by LMR
GNDREF
LM7805_LR
input
ground
output
U1
+9V
C1
0.33 uF : 25V
C2
0.1 uF : 25V
TP4
Incoming 5V test point
+5V
JP3
Internal 5V Jumper
PWR_FLAG
IR Sensor
R4
220k : 1/4W
R3
100k : 1/4W
Vout1
PWR_FLAG
JP7
IR Sensor 5V Jumper
+5V
C4
0.1 uF : 25V
TP6
PT_Test
Vref
RV1
10k
IR_LED
R2
220 : 1/4W
D1
TSAL6100_IR_LED_LMR
Q2
BPW96B_PhotoTransistor_LMR
R5
220k : 1/4W
R6
100k : 1/4W
Vout2
R7
2k : 1/4W
R8
2k : 1/4W
R9
1k : 1/4W
TP5
Vref_Test
MCP6004_OPAMP_LMR
U6
User Interface LEDs
RED_LED
R10
220 : 1/4W
J2
Red_LED_Screw_Terminal
BLUE_LED
R11
220 : 1/4W
J3
Blue_LED_Screw_Terminal
GREEN_LED
R1
220 : 1/4W
J4
Green_LED_Screw_Terminal
(Screw Terminals to reach User)
Curiosity Nano
U2
PIC_CuriosityNano_LMR
PIC18F57Q43_CuriosityNano_LR
U8
U7
Extra_Pins_Header
JP5
PIC 5V jumper
C3
0.1 uF : 25V
JP6
PIC Capacitor Jumper
MTR_ana_in
MTR_ana_out
SPKR_ana_in
SPKR_ana_out
WGT_ana_in
WGT_ana_out
Vout2
MTR_dig_in
MTR_dig_out
SPKR_dig_in
SPKR_dig_out
WGT_dig_in
WGT_dig_out
TEST_BUTTON
Test Points
TP1
Vout2_Test
TP2
Vout1_Test
Jumpers
JP8
5V_TO_9V_JUMPER
JP2
External 9V Jumper
External_9V
Debugging Button and LED
TEST_LED
R12
220 : 1/4W
D4
Red_Test_LED
SW1
TEST_Button
R13
10k : 1/4W
Connections to Subsystems
Weight Sensor Connector
2x4_IDC_Weight_LMR
U3
JP22
JP23
JP24
JP25
JP26
JP27
JP28
Speaker Connector
2x4_IDC_Int_IR_LMR
U4
JP13
JP16
JP17
JP18
JP19
JP20
JP21
Hinge Motor Connector
2x4_IDC_Motor_LMR
U5
JP9
JP1
JP12
JP15
JP14
JP10
JP11
External_9V
Lia Ryan
EGR 304 Team 202
Sheet: /
File: Subsystem_Schematic_Design_Lia_Ryan_EGR304_Team_202.kicad_sch
Title: Exterior IR Senser Subsystem Schematic: AutoCan
Size: A4
Date: 2025-10-20
Rev: 5
KiCad E.D.A. 9.0.4
Id: 1/1